

QUESTIONS:

Question 1 — Single Perceptron Implementation

Build a **Perceptron** in Python.

Dataset:

$X = [[0,0], [0,1], [1,0], [1,1]]$

$y = [0, 0, 0, 1]$

Tasks:

1. Initialize weights and bias randomly
 2. Implement perceptron function
 3. Apply activation function (step function)
 4. Train perceptron for at least 10 iterations
 5. Print final weights and predictions
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Question 2 — Logic Gate using Perceptron

Implement **AND, OR, and NOT gates** using a **Perceptron**.

Input Data:

- AND gate inputs: (0,0), (0,1), (1,0), (1,1)
- OR gate inputs same
- NOT gate inputs: (0), (1)

Tasks:

1. Define weights and bias for each gate
 2. Implement perceptron function
 3. Test all input combinations
 4. Display output for each gate
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Question 3 — TensorFlow Single Layer Perceptron

Use **TensorFlow** to build a neural network.

Tasks:

1. Generate dataset using `make_classification`
 2. Split dataset into train/test
 3. Standardize dataset
 4. Build model using:
 Input layer (2 features)
 Dense(1, activation='sigmoid')
 5. Compile model (Adam + binary_crossentropy)
 6. Train model for 50 epochs
 7. Evaluate accuracy
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Question 4 — PyTorch Simple Neural Network

Use **PyTorch** to build a binary classifier.

Tasks:

8. Create random dataset (1000 samples, 2 features)

9. Convert data into tensors
10. Build simple neural network:
 - Input layer → 2 neurons
 - Output layer → 1 neuron
11. Use sigmoid activation
12. Define loss function (BCELoss)
13. Train model for 50 epochs
14. Print loss after each epoch

Question 5 — Model Evaluation (TensorFlow)

Using trained **TensorFlow** model:

Tasks:

1. Predict test data outputs
2. Convert probabilities into binary values (0/1)
3. Generate confusion matrix
4. Print classification report
5. Calculate accuracy manually and using sklearn